

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

AMMS1623 CALCULUS AND ALGEBRA

THURSDAY, 9 OCTOBER 2025

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMMS1623 CALCULUS AND ALGEBRA**Question 1**

- a) Find an equation of the line that passes through $(0,0)$ and $(1,8)$ in the form of $ax + by + c = 0, a \geq 0$. (5 marks)
- b) Find the shortest distance between the line $l, 3x + 5y + 11 = 0$ and the point $O(0,0)$. Correct your final answer to 2 decimal places. (5 marks)
- c) Find an equation of the circle with radius 5 and centre $(1,3)$ in the expanded form $x^2 + y^2 + bx + dy + e = 0$. (5 marks)
- d) Find the angle between the vectors $\mathbf{u} = \langle 8,0,0 \rangle$ and $\mathbf{v} = \langle 1,1,1 \rangle$, rounded to 1 decimal place in degree. (5 marks)
- e) Find an equation of the plane in the form of

$$ax + by + cz + d = 0, a \geq 0,$$
that passes through the points $P(0,0,0)$, $Q(1,1,0)$ and $R(0,1,1)$. (5 marks)
- [Total: 25 marks]

Question 2

- a) Use Euclidean algorithm to find $d = \text{GCD}(56,24)$ and write $d = s24 + t56, s, t \in \mathbb{Z}$. Hence find $\text{LCM}(56,24)$. (8 marks)
- b) Solve the following system of equations
 $x \equiv 2 \pmod{5}$
 $x \equiv 5 \pmod{7}$ (7 marks)
- c) Differentiate each of the following functions. Simplify your answers.
- (i) $y = x^8 + 3x$ (2 marks)
- (ii) $y = (x^8 + 1)^8$ (4 marks)
- (iii) $y = \frac{x^3}{(x+3)^8}$ (4 marks)
- [Total: 25 marks]

AMMS1623 CALCULUS AND ALGEBRAQuestion 3

- a) Find all the relative extreme points and inflection point of the function $y = x^3 - 12x + 1$. Use the second derivative test to determine whether each of the relative extreme points is a relative maximum point or relative minimum point. (8 marks)
- b) Find the linear approximation of the function $f(x) = \sqrt{x+4}$ at $x = 0$ and use it to approximate the number $\sqrt{4.1}$. Correct your final answer to 3 decimal places. (5 marks)
- c) Find an equation of the tangent line to the curve with parametric equations, $x = t^5 + 1$, $y = t^3 + 2t$, at the point corresponding to the given value of the parameter $t = 2$ in the form of $ax + by + c = 0$, $a \geq 0$. (5 marks)
- d) An experimental rocket fired straight up into the air has height

$$s(t) = 300t - 3t^2$$
metres, after t seconds.
- (i) What is the rocket's initial velocity? (2 marks)
- (ii) What is the velocity after 50 seconds? (1 mark)
- (iii) What is the acceleration when $t = 8$? (2 marks)
- (iv) What is the maximum height of the rocket? (2 marks)
- [Total: 25 marks]

AMMS1623 CALCULUS AND ALGEBRA**Question 4**

- a) Evaluate the given indefinite integral
- (i) $\int x^3 + 1 \, dx$ (3 marks)
- (ii) $\int 2x(x^2 + 1)^8 \, dx$. (7 marks)
- b) Find the area under $y = x^2$ and above x -axis from $x = 0$ to $x = 3$. (5 marks)
- c) Determine the mean value of $f(x) = x^5$ over the interval from $x = 0$ to $x = 2$. Correct your final answer to 2 decimal places. (5 marks)
- d) Find the the volume of the solid of revolution generated by revolving about the x -axis the region under $y = 3x^2 + 1$ from $x = 0$ to $x = 5$. Correct your final answer to 2 decimal places. (5 marks)

[Total: 25 marks]

AMMS1623 CALCULUS AND ALGEBRA**Distance Formula**

The distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the plane is

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint Formula

The midpoint of the line segment from $A(x_1, y_1)$ to $B(x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Distance between a Point and a Line

The distance d from a point $Q(x_0, y_0)$ to the line $ax + by + c = 0$ is

$$d = \frac{|a(x_0) + b(y_0) + c|}{\sqrt{a^2 + b^2}}$$

Equation of a Circle

An equation of the circle with centre (h, k) and radius r is

$$(x - h)^2 + (y - k)^2 = r^2$$

Vectors in Two Dimensions

If $\mathbf{u} = \langle a_1, a_2 \rangle$ and $\mathbf{v} = \langle b_1, b_2 \rangle$, then their dot product is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2$$

If θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

The angle θ between $\mathbf{u}$ and $\mathbf{v}$ satisfies

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|}$$

Vectors in Three Dimensions

The distance between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by the

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in space, then their dot product is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

The Cross Product of Vectors in Space

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in space, then their cross product is the vector

$$\mathbf{u} \times \mathbf{v} = (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

We can calculate the cross product using determinants.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The scalar triple product

The product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is called the **scalar triple product** of the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. You can check that the scalar triple product can be written as the following determinant:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

AMMS1623 CALCULUS AND ALGEBRA**Equations of Lines and Planes**

A **line** passing through the point $P(x_0, y_0, z_0)$ and parallel to the vector $\mathbf{v} = \langle a, b, c \rangle$ is described by the **parametric equations**

$$x = x_0 + at, y = y_0 + bt, z = z_0 + ct$$

where t is any real number.

A **plane** containing the point $P(x_0, y_0, z_0)$ and having the normal vector $\mathbf{n} = \langle a, b, c \rangle$ is described by the **plane equation**

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

The Distance from a Point to a Line

Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

The Distance from a Point to a Plane

If P is a point on a plane with normal $\mathbf{n}$, then the distance from any point S to the plane is

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$$

where $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is normal to the plane.

Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

Quotient Rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

The Chain Rule

To differentiate $f(g(x))$, differentiate first the outside function $f(x)$ and substitute $g(x)$ for x in the result. Then, multiply by the derivative of the inside function $g(x)$. Symbolically,

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x).$$

Differentiation Formula	Corresponding Integration Formula
$\frac{d}{dx}(x^r) = rx^{r-1}$	$\int rx^{r-1}dx = x^r + C$ or
$\frac{d}{dx} \left(\frac{x^{r+1}}{r+1} \right) = x^r$	$\int x^r dx = \frac{x^{r+1}}{r+1} + C, r \neq -1$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\int \frac{1}{x} dx = \ln x + C$
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}(\cos x) = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$

AMMS1623 CALCULUS AND ALGEBRA**Integration by Substitution**

$$\int f(g(x))g'(x)dx = F(g(x)) + C$$

Integration by Parts

$$\int f(x)g(x)dx = f(x)G(x) - \int f'(x)G(x)dx$$

Linear Approximation

$$f(x) \approx L(x) = f(a) + f'(a)(x - a)$$

The **mean value** of $f(x)$ over the interval from $x = a$ to $x = b$.

$$\frac{1}{b-a} \int_a^b f(x)dx$$

The **volume of the solid** of revolution generated by revolving the region under $y = g(x)$ from $x = a$ to $x = b$ about the x -axis.

$$\int_a^b \pi[g(x)]^2 dx$$

Quadratic Formula

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$